



TULSIRAMJI GAIKWAD-PATIL College of Engineering and Technology

Wardha Road, Nagpur - 441108

Accredited with NAAC A+ Grade

Approved by AICTE, New Delhi, Govt. of Maharashtra



(An Autonomous Institute Affiliated to RTM Nagpur University with NAAC A+ Grade)

Department of Information Technology

Session 2024-25(Odd Semester)

VISION	MISSION
To emerge as a learning hub and center of excellence in the domain of Information Technology	[M1] To impart quality technical education through effective teaching learning process. [M2] To provide a platform to address societal issues as well as challenges faced by IT industries. [M3] To foster a culture of research and impart innovative and entrepreneurial skills in the field of IT. [M4] To ensure overall development of students and staff by inculcating knowledge and professional ethics as a part of lifelong learning.

END SEMESTER

Semester: V

Programme: B. Tech Information Technology

Course: Design & Analysis of Algorithms

Time: 3 Hours

Max Marks: 60 Marks

Instructions to candidates:

1. All questions are compulsory in Group A
2. Solve any two sub-question out of three in Group B
3. Assume suitable data wherever necessary
4. All question carries marks as indicated

BIT33503	Course Outcome
CO1	To understand the foundational concepts, characteristics, and performance analysis of algorithms.
CO2	To explore algorithm design paradigms.
CO3	To classify classical algorithms and recurrence relations.
CO4	To develop efficient solutions using structured techniques.
CO5	To understand complexity classes and non-deterministic algorithms.

Group A				
Que. 1	Answer the following Question	CO	BTL	Marks
a.	Define Big-O notation.	CO1	2	2
b.	What is Merge Sort?	CO2	2	2
c.	Define overlapping subproblems.	CO3	2	2
d.	What is graph coloring?	CO4	3	2
e.	Define NP-Complete.	CO5	2	2
Group B				
Que. 2	Answer the following Question (Solve any Two)	CO	BTL	Marks

a.	Solve recurrence using Master theorem.	CO1	2	5
b.	Explain algorithm characteristics.	CO1	2	5
c.	Explain asymptotic notations.	CO1	2	5
Que 3.	Answer the following Question (Solve any Two)	CO	BTL	Marks
a.	Explain Quick Sort with complexity.	CO2	2	5
b.	Explain Prim's algorithm.	CO2	3	5
c.	Explain Fractional Knapsack.	CO2	3	5
Que 4.	Answer the following Question (Solve any Two)	CO	BTL	Marks
a.	Explain DP vs Greedy	CO3	2	5
b.	Solve Matrix Chain Multiplication.	CO3	3	5
c.	Explain Floyd Warshall algorithm.	CO3	4	5
Que 5.	Answer the following Question (Solve any Two)	CO	BTL	Marks
a.	Explain N-Queen problem.	CO4	2	5
b.	Explain Sum of Subsets.	CO4	4	5
c.	Explain Branch and Bound.	CO4	3	5
Que 6.	Answer the following Question (Solve any Two)	CO	BTL	Marks
a.	Explain NP-Complete problems.	CO5	2	5
b.	Discuss Cook's theorem.	CO5	4	5
c.	Explain complexity classes.	CO5	3	5

Bloom's Taxonomy Level (BTL)- Remember , Understand , Apply, Analyze , Evaluate , Create